Roma Tre

Information Visualization

Task modeling

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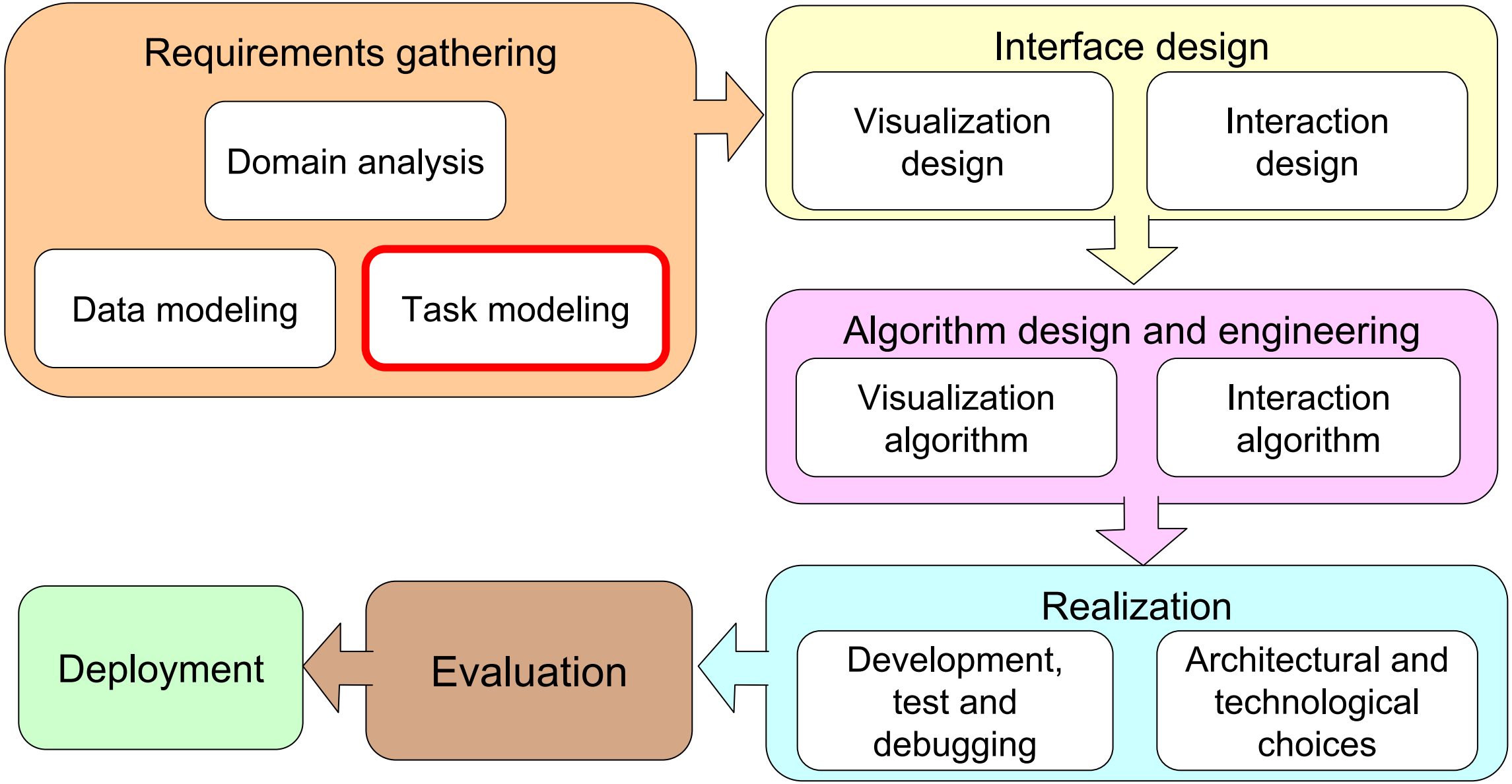
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A taxonomy of task taxonomies

- Historical task taxonomies
  - low level task taxonomy
    - [Wehrend & Lewis 90]
  - task by data type taxonomy
    - [Shneiderman 96]
  - Andrienko task framework
    - [Andrienko & Andrienko 06]
- Contemporary task taxonomies
  - analytic task taxonomy
    - [Amar et al., 05]
  - graph oriented task taxonomy
    - [Lee et al. 06]
  - task taxonomy for temporal graphs
    - [Kerracher et al 15]
  - health data visualization task taxonomy
    - [Theis et al 18]

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Design pipeline



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Historical task taxonomies

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Low level task taxonomy

- [Wehrend & Lewis 90] created a “low-level”, domain-independent task taxonomy of visualization environments
  - the purpose was that of classifying visualization techniques based on the supported tasks
- Eleven basic actions
  - identify, locate, distinguish, categorize, cluster, distribution, rank, compare within relations, compare between relations, associate, correlate

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TTT: Task by data type taxonomy

- [Shneiderman 96] proposed a task taxonomy dependent on the data type

Data type	Task
1D (Linear: text, lists, …)	Overview
2D (Planar or maps)	Zoom
3D (Volumes and surfaces)	Filter
Temporal	Details-on-demand
Multidimensional	Relate
Tree	History
Network	Extract

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Shneiderman high level tasks

- Overview
  - gain an overview of the entire collection → 1D (linear: text, lists, …)
- Zoom
  - zoom in on items of interest → 2D (planar or maps)
- Filter
  - filter out uninteresting items → 3D (volumes and surfaces)
- Details-on-demand
  - select an item or group and get details when needed → temporal
- Relate
  - view relations hips among items → multidimensional
- History
  - keep a history of actions to support undo, replay, and progressive refinement → tree?
- Extract
  - allow extraction of sub-collections and of the query parameters → network?

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Andrienko Task Framework (ATF)

- Introduced by [Andrienko & Andrienko 06]
  - see also [Kerracher et al 15]
- Classification of tasks
  - lookup tasks
    - direct lookup
      - searching for a characteristic given a reference
    - inverse lookup
      - searching for a reference given a characteristic
  - comparison tasks
    - direct comparison
      - the task target is the relation between characteristics or patterns
    - reverse comparison
      - the task target is the relation between references or reference sets
  - relation seeking
      - the opposite of comparison, where the target is the components associated with a given relation

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Task type		Target	Constraint	Elementary (example)	Synoptic (example)
Lookup	Direct	caracteristic	referential	Which department did Author A belong to in 2012?	What is the trend in Author A's publication counts 2012-2014?
	Inverse	referential	characteristic	Which author(s) had more than 4 publications in any year?	Find authors who move frequently between departments.
Comparison	Direct	relation	characteristic	Compare the publication counts of Authors A and B in 2014.	Compare the trend in Author A's publication counts for 2012-2014 with the trend for 2009-2012.
	Inverse	relation	referential	Did Author A's highest publishing count occur before or after his lowest?	Compare the time periods over which Author A's publication counts were increasing with the time periods over which they were decreasing.
Relation seeking		characteristic/ referential	relation	Find the year in which Author B moved departments (i.e. consecutive years where Author B belonged to two different departments).	Find authors with similar patterns in movement between departments.

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Analytic task taxonomy

- Approach and metodology [Amar et al., 05]
  - ask users to produce simple questions about five datasets
  - classify the questions clustering them by similarity
- Chosen datasets:

Data Set	Dimensionality	Cardinality	Questions
Cereals	15	78	43
Mutual funds	14	987	14
Cars	10	407	53
Films	10	1742	47
Grocery surveys	8	5164	39

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Contemporary task taxonomies

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Analytic task taxonomy [Amar et al., 05]

- Retrieve value
- Find extremum
- Determine range
- Compute derived value

value oriented

- Filter
- Sort
- Cluster
- Find anomalies

perspective change

- Characterize distribution
- Correlate

high level

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Retrieve value & Find extremum

- Retrieve value
  - given a set of specific cases, find attributes of those cases
  - examples
    - What is the mileage per gallon of the Audi TT?
    - How long is the movie “Gone with the Wind”?
- Find extremum
  - find data cases possessing an extreme value of an attribute over its range within the data set
  - examples
    - What is the car with the highest MPG?
    - What director/film has won the most awards?
    - What Robin Williams film has the most recent release date?

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Determine range & Compute derived value

- Determine range
  - given a set of data cases and an attribute of interest, find the span of values within the set
  - examples
    - What is the range of film lengths?
    - What is the range of car horsepowers?
- Compute derived value
  - given a set of data cases, compute an aggregate numeric representation of those data cases
  - examples
    - What is the gross income of all stores combined?
    - How many manufacturers of cars are there?
    - What is the average calorie content of Post cereals?

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Filter & Sort

- Filter
  - given some concrete conditions on attribute values, find data cases satisfying those conditions
  - examples
    - What Kellogg's cereals have high fiber?
    - What comedies have won awards?
    - Which funds underperformed the SP-500?
    - What actresses are in the data set?
- Sort
  - given a set of data cases, rank them according to some ordinal metric
  - examples
    - Order the cars by weight
    - Rank the cereals by calories

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Cluster & Find anomalies

- Cluster
  - given a set of data cases, find clusters of similar attribute values
  - examples
    - Are there groups of cereals w/ similar fat/calories/sugar?
    - Is there a cluster of typical film lengths?
- Find anomalies
  - identify any anomalies within a given set of data cases with respect to a given relationship or expectation, e.g. statistical outliers
  - examples
    - Are there any outliers in protein?
    - Are there exceptions to the relationship between horsepower and acceleration?

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Characterize distribution & Correlate

- Characterize distribution
  - given a set of data cases and a quantitative attribute of interest, characterize the distribution of that attribute's values over the set
  - examples
    - What is the distribution of carbohydrates in cereals?
    - What is the age distribution of shoppers?
- Correlate
  - given a set of data cases and two attributes, determine useful relationships between the values of those attributes
  - examples
    - Is there a correlation between carbohydrates and fat?
    - Do different genders have a preferred payment method?
    - Is there a trend of increasing film length over the years?

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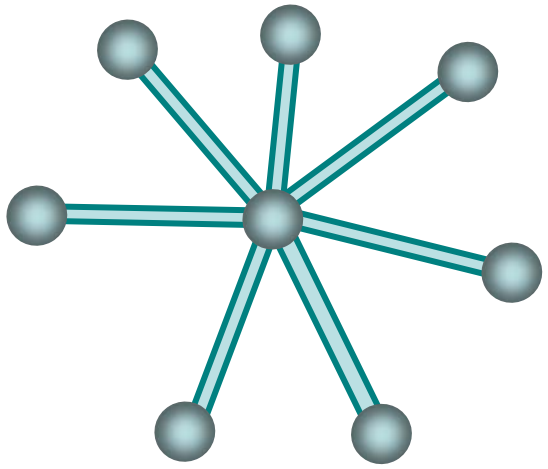
Graph-oriented tasks

- To the low-level tasks just seen several tasks are exclusive of graphs [Lee et al. 06]
  - topology-based tasks
    - adjacency
    - reachability
    - commmon Connection
    - connectivity
  - attribute-based tasks
  - browsing tasks
  - overview tasks

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Adjacency

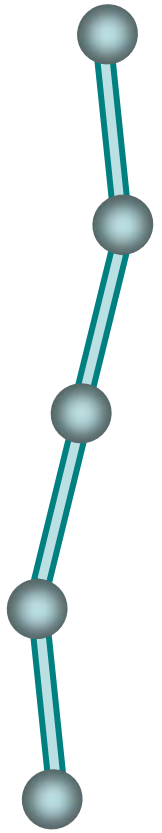
- Adjacency (direct connection among nodes)
  - find the set of nodes adjacent to a specific node
    - Who are the friends of Eric?
  - count nodes that are adjacent to a specific node
    - How many kinds of organisms do golden eagles eat?
  - find the node that has the highest degree
    - Who is the most popular person?



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Reachability

- Reachability (direct or indirect connection among nodes)
  - find the set of nodes reachable from a node
    - Who are your friends, your friends' friends, and so on?
  - how many nodes are reachable from a node?
    - How many friends are you connected to in this way?
  - find the set of nodes reachable from a node where the distance is less than or equal to n
    - To what cities can we go from Seoul, Korea by changing planes only once?
  - how many nodes are reachable from a node where the distance is less than or equal to n?



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Commmon connection and connectivity

- Common connection
  - given nodes, find a set of nodes that are connected to all of them
    - Find all the people who know both John and Jack
- Connectivity
  - find the shortest path between two nodes
    - What is the shortest path from Seoul, Korea to Athens, Greece?
  - identify clusters
    - Count the number of clusters in a social network
  - identify connected components
  - find bridges
    - What link disconnects the graph if removed?
  - find articulation points
    - What node disconnects the graph if removed?

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Attribute-based tasks

- On the Nodes
  - find the nodes having a specific attribute value
    - Who are the teachers among the people currently shown on screen?
    - How many engineers are there among the ones currently shown on screen?
  - review the set of nodes
    - Are there any foreign-sounding names?
- On the Links
  - given a node, find the nodes connected only by certain types of links
    - Find the nodes connected by “is-a” relationships from the “Biological Process” node
  - which node is connected by a link having the largest/smallest value?
    - If a link has an attribute representing the percentage of the diet, what is main food of American crow?

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Browsing tasks

- Follow Path
  - follow a given path
    - Follow a chain of friends in a social network
    - Follow the flow of energy from grasses, to a rabbit that eats grass, to a carnivore that eats the rabbit, and to a carnivore that eats that carnivore
- Revisit
  - return to a previously visited node
    - Explore another friend of Eric
    - Find another carnivore that eats the same rabbit

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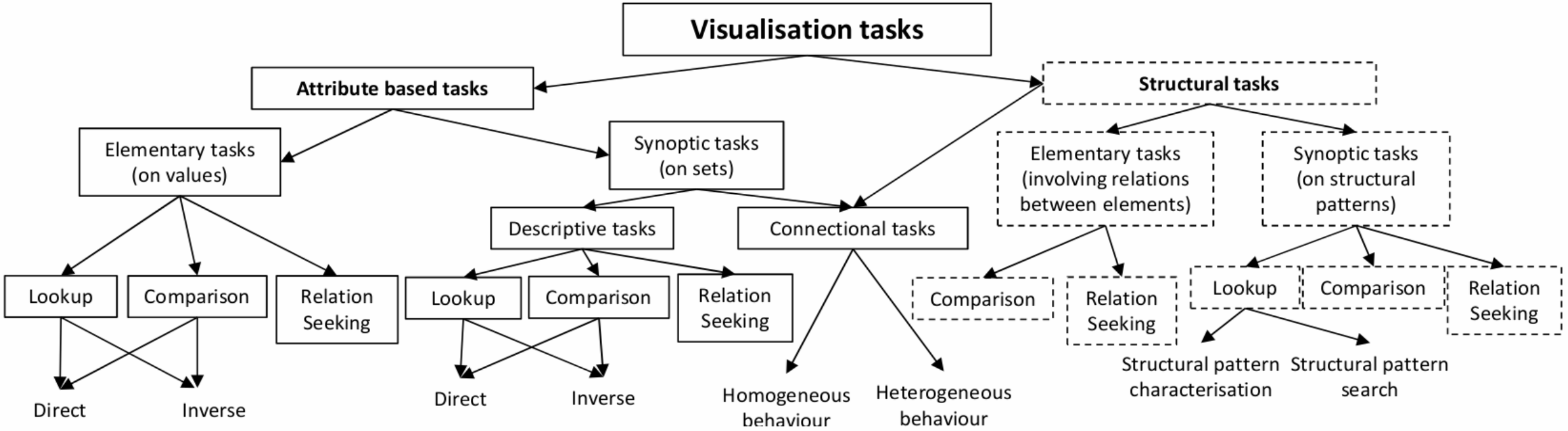
Overview tasks

- These are high-level tasks that ask for a quick and approximate answer
  - estimate size of the network
  - estimate the number of connected components
  - is the network clustered?
  - can you identify different patterns of connection?
  - has the network a small-world structure?

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Task taxonomy for temporal graphs

- Described in [Kerracher et al 15]
- Extends the classification of [Andrienko & Andrienko 06]



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Taxonomy for Health Data Visualization

- Described by [Theis et al 18]
- Addresses both task and data models
- Based on the answers given by 98 participants of an online survey
  - 51 older adults and 47 eHealth experts
- Distinguishes between high level targets as consultation, diagnosis, mentoring, and monitoring
- Also includes editing tasks

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An excerpt of Table 3 from [Theis et al 18]

Visualization task	N	Adults, n (%)	experts, n (%)	χ2	P value
Consultation					
Perceive information	53	39 (74)	14 (26)	0.3	.75
Search information	41	27 (66)	14 (34)	4.6	.05
Record information	41	31 (76)	10 (24)	0.1	.88
Present information	40	27 (68)	13 (33)	2.9	.15
Annotate information	40	29 (73)	11 (28)	0.3	.78
Query information	39	29 (74)	10 (26)	0.1	.88
Diagnosis					
Perceive information	47	34 (72)	13 (28)	0.6	.55
Discover information	47	34 (72)	13 (28)	0.6	.45
Search information	46	33 (2)	13 (28)	0.8	.37
Locate information	43	33 (77)	10 (23)	0.2	.66
Identify information	42	34 (81)	8 (19)	2.0	.15

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Final remarks

- If the purposes of visualization were fully known in advance visualization would not be needed
  - an algorithm would perform those task better than a visualization system
    - why visualizing data if you simply need the answer to the question: “What comedies have won awards?”
- If visualization is needed, this is because the user aims are, at the moment, uncertain
  - however, we know that user aims can be carried out by leveraging some basic tasks
  - by combining these tasks the users answer their questions

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